

→ Partnership

$A : B = 2 : 3$

$B : C = 5 : 7$

$A : B : C = \boxed{}$

Partnership

Anand and Deepak started a business investing Rs. 22500 and Rs. 35000 respectively. Out of a total profit of Rs. 13800, Deepak's share is

- (a) Rs. 5400
- (b) Rs. 7200
- (c) Rs. 8400
- (d) Rs. 9600

$$A : D = 22500 : 35000 = 9 : 14 \rightarrow 23$$

$$D = \frac{14}{23} \times 13800 = 8400$$

Nikita and Nishita enters into a partnership by invest investing Rs.50,000 and Rs.40,000, respectively. They agreed to share profits in the ratio of their capitals. Find out the share of Nikita when profit of the business is Rs.22500 after a year.

- (a) Rs1,500
- (b) Rs9,500
- (c) Rs10,500
- (d) None of these

$$\frac{Nik}{Nish} = \frac{5}{4} \rightarrow 9$$

$$Nik = \frac{5}{9} \times 22500 = 12500$$

P, Q and R invested Rs. 45000, Rs. 70000 and Rs. 90000 respectively to start a business. At the end of 2 years, they earned a profit of Rs. 164000. What will be Q's share in the profit?

- (a) Rs 36000
(b) Rs 56000
(c) Rs 64000
(d) Rs 72000
(e) None of these

$$P : Q : R = 9 : 14 : 18 \rightarrow \underline{\underline{41}}$$

$$Q = \frac{14}{41} \times 1,64,000 = \underline{\underline{56,000}}$$

Rahul, Arun and Sumit started a business. Rahul Invested $\frac{1}{2}$ part, Arun $\frac{1}{3}$ part and rest of the capital was invested by Sumit. The ratio of their profits will be

- (a) 2 : 3 : 1
(b) 3 : 2 : 1
(c) 2 : 3 : 6
(d) 3 : 2 : 5

$$\text{Total} = 1$$

$$R : A : S = \frac{1}{2} : \frac{1}{3} : \frac{1}{6}$$

$$R = \frac{1}{2} \quad A = \frac{1}{3} \quad \left. \vphantom{\frac{1}{2}} \right) + = \frac{5}{6}$$

$$= \frac{6}{2} : \frac{6}{3} : \frac{6}{6}$$

$$= \underline{\underline{3 : 2 : 1}}$$

$$S = 1 - \frac{5}{6} = \frac{1}{6}$$



P and Q started a business investing Rs. 85000 and Rs. 15000 respectively. In what ratio the profit earned after 2 years be divided between P and Q respectively?

- (a) 3 : 4
(b) 3 : 5
(c) 15 : 23
(d) 17 : 23
(e) None of these

$$P : Q = \frac{85000}{17} : \frac{15000}{3}$$

$$= \underline{\underline{17 : 3}}$$

Mahesh, Suresh and Ganesh entered into a partnership. partnership business. Mahesh invested Rs.16,000 for 9 months. Suresh invested Rs.12,000 for 6 months and Ganesh invested Rs.8,000 for 12 months. At the end of a year, there was a profit of Rs.26,000. Find out the share of Suresh in the profit.

- (a) Rs.8,000
(b) Rs.7,500
(c) Rs.6,000
(d) None of these

$$M : S : G = \overset{2}{\cancel{16000}} \times \overset{3}{\cancel{9}} : \overset{3}{\cancel{12000}} \times \overset{2}{\cancel{6}} : \overset{2}{\cancel{8000}} \times \overset{4}{\cancel{12}}$$

$$= \underline{6 : 3 : 4} \rightarrow \underline{13}$$

$$S = \frac{3}{13} \times \frac{26000}{1} = \underline{\underline{6000}}$$

Samaira, Mahira and Kiara rented a set of DVDs at a rent of Rs. 578. If they used it for 8 hours, 12 hours and 14 hours respectively, what is Kiara's share of rent to be paid?

- (a) Rs 192
(b) Rs 204
(c) Rs 215
(d) Rs 238
(e) None of these

$$S : M : K = \overset{4}{\cancel{8}} : \overset{6}{\cancel{12}} : \overset{7}{\cancel{14}} = \underline{4 : 6 : 7} \rightarrow \underline{17}$$

$$K = \frac{7}{17} \times 578 = \underline{\underline{238}}$$

Three persons started a placement business with a capital of Rs. 3000. B invests Rs. 600 less than A and C invests Rs. 300 less than B. What is B's share in a profit of Rs. 886?

- (a) Rs 443
(b) Rs 354.40
(c) Rs 265.80
(d) Rs 177.20

Handwritten solution for the first problem:

$$A : B : C = 5 : 3 : 2$$

$$A + B + C = 3000$$

$$B = A - 600$$

$$C = B - 300 = (A - 600) - 300 = A - 900$$

$$A + (A - 600) + (A - 900) = 3000$$

$$3A - 1500 = 3000$$

$$3A = 4500$$

$$A = 1500$$

$$B = \frac{3}{5} \times 1500 = 900$$

$$B = 265.8$$

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Sita and Gita enter into a partnership. Sita contributes Rs. 5,000 while Gita contributes Rs. 4,000. After 1 month, Gita withdraws $\frac{1}{4}$ part of her contribution and after 3 months Sita invests Rs. 2,000 more. When Gita withdraws her investment, at the same time, Rita joins them by investing Rs. 7,000. If at the end of 1 year there is a profit of Rs. 1,218, what will be share of Rita in the profit?

- (a) Rs. 488.47
(b) 8477.37
(c) 588.47
(d) None of these

$$S = 5000 \times 12 + 2000 \times 9 = 78000$$

$$G = 4000 \times 1 + 3000 \times 11 = 37000$$

$$R = 7000 \times 11 = 77000$$

$$S : G : R = 78 : 37 : 77$$

$$R = \frac{77}{192} \times 1218 =$$



A starts business with an investment of Rs3500. Five months later B joins A as a partner. After a year, the profits are divided in the ratio of 2:3. How much did B contribute?

- (a) Rs7,000
(b) Rs11,000
(c) Rs9,000
(d) None of these

$$\frac{A}{B} = \frac{2}{3} = \frac{3500 \times 12}{x \times 7}$$



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$$x = \frac{3500 \times 12 \times 3}{2 \times 7} = 9000$$



Gupta and Bansal enters into a partnership with their capitals in the ratio 5:6. At the end of 8 months, Gupta withdraws his capital. If they receive their shares profits in the ratio of 5:9, find out how long Bansal's capital was invested in the business?

- (a) 10 months
(b) 12 months
(c) 14 months
(d) None of these

$$\frac{G}{B} = \frac{5 \times 8}{6 \times x} = \frac{5}{9}$$

$$9C = 12 \quad \checkmark$$

A, B and C enter into a partnership. A invests some money at the beginning, B invests double the amount after six months and C invests thrice the amount after eight months. If the annual profit be Rs 27000; C's share (in Rs) is

- (a) Rs 8625
(b) Rs 9000
(c) Rs 10800
(d) Rs 11250

$$\begin{aligned} A &= x \times 12 = 12x \\ B &= 2x \times 6 = 12x \\ C &= 3x \times 4 = 12x \end{aligned}$$

$$\rightarrow A:B:C = 1:1:1$$

Arun, Kamal and Vinay invested Rs 8000, Rs 4000 and Rs 8000 respectively in a business. Arun left after six months. If after eight months, there was a gain of Rs 4005, then what will be the share of Kamal?

- (a) Rs 890
(b) Rs 1335
(c) Rs 1602
(d) Rs 1780

$$\begin{aligned} A &= 8000 \times 6 \\ K &= 4000 \times 8 \\ V &= 8000 \times 4 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} A : K : V = 3 : 2 : 1 \rightarrow 9$$

$$K = \frac{2}{9} \times 4005 = 890$$

Anju and Brijesh enters into a partnership with their capitals in the ratio of 5:9. At the end of 8 months, Anju withdraws her capital. If they receive their share of profit in the ratio of 4:9, find out how long Brijesh's capital was invested in the business.

- (a) 8 months
(b) 10 months
(c) 12 months
(d) None of these

$$\frac{A}{B} = \frac{5 \times 8^2}{9 \times x} = \frac{4}{9}$$

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$$x = 10 \downarrow$$

A and B entered into a partnership investing Rs 16000 and Rs 12000 respectively. After 3 months, A withdrew Rs 5000 while B invested Rs 5000 more. After 3 more months, C joins the business with a capital of Rs 21000. The share of B exceeds that of C, out of a total profit of Rs 26,400 after one year by

- (a) Rs 2400
(b) Rs 3000
(c) Rs 3600
(d) Rs 4800

$$\begin{aligned} A &= 16000 \times 3 + 11000 \times 9 \\ B &= 12000 \times 3 + 17000 \times 9 \\ C &= 21000 \times 6 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} A : B : C = 147 : 189 : 126$$

$$A : B : C = 7 : 9 : 6$$

$$= \frac{3}{22} \times 26,400 = 3600$$



Aman started a business investing Rs 70000. Rakhi joined him after six months with an amount of Rs 105000 and Sagar joined them with Rs 1.4 lakhs after another six months. The amount of profit earned should be distributed in what ratio among Aman, Rakhi and Sagar respectively, 3 years after Aman started the business?

- (a) 7 : 6 : 10
- (b) 12 : 15 : 16
- (c) 42 : 45 : 56
- (d) Cannot be determined
- (e) None of these

$$\begin{aligned}
 A &= \frac{70000}{100000} \times \frac{36}{6} \\
 R &= \frac{105000}{100000} \times \frac{30}{6} \\
 S &= \frac{140000}{100000} \times \frac{24}{6}
 \end{aligned}$$

$$= 12 : 15 : 16$$